

CY 642: Molecular- and Bio-electronics

Course Code	CY 642
Course Name	Molecular- and Bio-electronics
L-T-P-C	3-0-0-3
Prerequisites	IC 130 and IC 241 /or permission from instructor
Intended for	B.Tech. (All branches), M.S. and Ph.D.
Distribution	Elective
Approval	5th Senate (Oct 2013)

Course Contents

- **Molecular electronics**
Moore's laws and beyond; metallic atom-size contacts; transport through molecular junctions, plastic electronics; liquid crystal devices; single electron devices; logic gates; device fabrication strategies (in detail); tools for molecular electronics. [8 Lectures]
- **Organic electronics**
Pi-conjugated molecules; electroactive organic compounds; organic semiconductors; structure- electronic property relationships; organic electronic devices (OLED, OPV, OPD, OTFT, OFET); gate dielectrics; flexible polymeric substrates; printed organic electronics; novel processing techniques; device performance and characterization. [8 Lectures]
- **Nanoelectronics**
Nanowires, nanotubes and nanostructures synthesis, characterization and uses (in brief); nanowire integrated circuitry; nanostructure enabled chemical sensing; supra molecular bioelectronics nanostructures; quantum dot devices; electromechanical actuators. [8 Lectures]
- **Bioelectronics**
Introduction; electron transfer through proteins; biosensors and biofuel cells; enzyme electrodes; electrochemical DNA sensors; biomolecules-semiconductor interfaces for sensing and detection; bio-nano hybrid systems for electronic devices; DNA- templated electronics; S-layer proteins in bioelectronics; computing with nucleic acids. [10 Lectures]
- **Current research** Review of recent literature through high-impact journal articles on relevant topics; guest Lecturers (2-3) by eminent scientists/professors from abroad through video conference (using NKN facilities); few simple hands-on experiments will be demonstrated/performed. [8 Lectures]

References

1. M. C. Petty, **Molecular Electronics: From Principles to Practice**, John Wiley & Sons, 2008.
2. J. C. Cuevas, E. Scheer, **Molecular Electronics: An Introduction to Theory and Experiment**, World Scientific Publishing Company, 2010.
3. H. Klauk, **Organic Electronics: Materials, Manufacturing, and Applications**, Wiley-VCH, 2006.
4. H. Klauk, **Organic Electronics II**, Wiley-VCH, 2006.
5. K. Iniewski, **Nanoelectronics: Nanowires, Molecular Electronics, and Nanodevices**, McGraw-Hill, 2010.
6. I. Willner, E. Katz, **Bioelectronics: From theory to applications**, Wiley-VCH, 2005.